

ChromaXR: Enhancing Color Perception for Users with Color Vision Deficiency

Rafael A. Goiás*
AKCIT
Federal University of Goiás

Lucas L. Neves†
AKCIT
Federal University of Goiás

Paulo Vitor S. Silva‡
AKCIT
Federal University of Goiás

Gustavo H. W. Barbosa§
AKCIT

Davi D. Freitas||
AKCIT
Federal University of Rio Grande do Norte

Rafael T. Sousa||
AKCIT
Federal University of Mato Grosso

Arlindo R. Galvão Filho**
AKCIT
Federal University of Goiás

ABSTRACT

This demo introduces an extended reality (XR) system designed to enhance environmental perception for individuals with Color Vision Deficiency (CVD). The system leverages a hybrid workflow: first, users perform a digital implementation of the Farnsworth D-15 Arrangement Test within a fully immersive Virtual Reality (VR) environment to ensure controlled lighting and diagnostic precision. Based on this assessment, the system generates a personalized visual profile, which is then applied as a tailored, static compensation filter in an Augmented Reality (AR) setting. This overlay optimizes color contrast and discrimination in the user's real-world field of view. At the demonstration venue, participants can experience the full XR pipeline, transitioning from VR-based calibration to AR-assisted vision, showcasing the potential of cross-reality tools as scientifically-grounded assistive technologies.

Index Terms: Augmented Reality, Color Vision Deficiency, Assistive Technology.

1 INTRODUCTION

Color Vision Deficiency (CVD) affects a significant segment of the population, limiting the ability to distinguish between certain hues. This impairment not only complicates everyday tasks, such as interpreting traffic signals or identifying color-coded items, but can also give rise to social discomfort and misunderstandings [2, 1].

Although accessibility tools exist in modern operating systems, these often rely on generic and static profiles (e.g., "Colorblind Mode"), disregarding individual variability in severity and chromatic confusion axis.

Virtual Reality (VR) offers a new paradigm for addressing this challenge, enabling both precise stimulus control for diagnosis and real-time intervention in scene rendering. This work proposes a distributed system architecture that integrates clinical psychophysics with computer graphics to deliver a personalized accessibility solution.

Unlike traditional approaches requiring high computational overhead for complete biological simulation (daltonization via matrix convolution), our approach implements a lightweight and efficient Parametric Chromatic Compensation strategy. The system employs a digitized Farnsworth-Munsell Dichotomous D-15

test [3] in VR to map the user's perceptual topology.

In this context, we propose a novel approach¹ that employs the vector algorithm proposed by Vingrys and King-Smith [5] to extract precise quantitative metrics (i.e., the Confusion Angle and the Confusion Index) and translates the diagnosis directly into HSL (Hue, Saturation, Lightness) color space. Resulting in a case-specific overlay filter, trigonometrically computed to maximize visual contrast perpendicular to the user's confusion axis, ensuring performance compatible with latency requirements of mobile XR applications.

2 SYSTEM ARCHITECTURE

The system follows a two-stage pipeline: a VR-based **Psychophysical Diagnostic** module that collects user interaction data from a digital Farnsworth D-15 test, and an Augmented Reality (AR)-based **chromatic correction** module that applies a personalized visual filter in real time. User color selections are transmitted to a server, which computes diagnostic parameters and returns a custom filter applied as a static overlay to the AR view, as demonstrated in Figure 1.

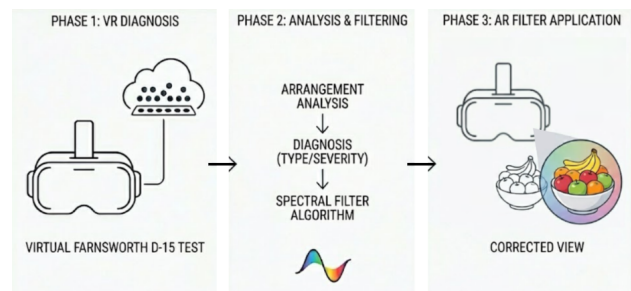


Figure 1: System pipeline

2.1 Psychophysical Diagnostic Module

This module implements the quantitative method of Vingrys and King-Smith [5] to convert the cap ordering topology into continuous variables.

First, the system processes the user's arrangement as an input vector $O = [c_0, c_1, \dots, c_{15}]$, where each cap index c_i is associated with a fixed chromatic coordinate (u'_i, v'_i) in the CIE 1976 UCS space, as defined by Farnsworth [3]. Following cap identification via Euclidean distance in the RGB space, the module proceeds to the **Moment of Inertia Calculation**. In this stage, the algorithm analyzes the chromatic difference vectors between adjacent caps in the user's sequence to derive the inertia tensor components (M_{uu}, M_{vv}, M_{uv}) [5].

¹Online Demonstration: <https://youtu.be/m1w1zWSj10s>

*e-mail: rafael.goiás@discente.ufg.br

†e-mail: lucas.neves@discente.ufg.br

‡e-mail: paulosantana@discente.ufg.br

§e-mail: guga.webster@gmail.com

||e-mail: davi.duarte.055@ufrn.edu.br

||e-mail: rafaelsousa@ufmt.br

**e-mail: arlindogalvao@ufg.br

The **Confusion Axis Angle** (θ) is derived from the moments of inertia and represents the principal orientation of error dispersion in the chromatic plane, serving as the main metric for classifying the deficiency type (Protan, Deutan, or Tritan).

Finally, the **Severity** is computed as a normalized function of the Confusion Index (C-index), obtained from the inertia matrix eigenvalues, where values ≤ 1.0 indicate normal vision. This maps the degree of visual loss onto a $[0, 1]$ scale, forming the digital profile used for AR compensation.

2.2 Adaptive Overlay Chromatic Correction Module

Based on the diagnostic parameters, the system generates a visual correction filter applied as an overlay on the Augmented Reality feed, using chromatic contrast enhancement perpendicular to the user's confusion axis.

The angle θ defines the direction of reduced chromatic discrimination, and the filter introduces a **Hue** orthogonal to this axis, increasing the perceptual salience of otherwise indistinguishable colors.

This transformation positions the filter color perpendicular to the confusion axis. The filter is defined in HSL space with the following parameters: **Hue (H)**: Derived from the confusion angle as per the equation above; **Saturation (S)**: Modulated by severity: $S = S_{base} \times (0.5 + 0.5 \times severity)$; **Lightness (L)**: Fixed value of 0.6 to maintain legibility.

Filter opacity (α) is set proportionally to condition severity, ranging from 30 (mild) to 180 (severe) on a 0–255 scale, while users with normal chromatic vision (C-index ≤ 1.0) receive a neutral filter ($\alpha = 0$). The resulting filter is encoded as an 8-character RGBA hexadecimal string and transmitted via TCP to be applied in real time as an overlay on the device's passthrough camera feed.

3 DEMONSTRATION

This demonstration showcases a cross-reality workflow on the Meta Quest 3, transitioning from a VR-based diagnosis to AR-based assistance, delivering highly personalized color enhancement for users with CVD.

3.1 Phase 1: VR Diagnostic

In the initial stage, the user is immersed in a distraction-free VR environment with a black background to standardize lighting and eliminate chromatic interference. A digital version of the Farnsworth D-15 Arrangement Test is then administered, in which the participant arranges colored caps into the correct chromatic sequence using the headset's spatial tracking, as shown in Figure 2. This procedure enables an accurate clinical assessment of the type and severity of color vision deficiency.



Figure 2: The Farnsworth D-15 Arrangement Test interface within the controlled VR environment.

3.2 Phase 2: AR Correction Application

In the second stage, the computed compensation filter is transmitted to the Meta Quest 3 and applied as a post-processing layer over

the high-resolution passthrough. The user then transitions to an AR view of the physical environment, where the personalized static filter enhances color contrast and discrimination in real time (Figure 3). This enables the user to perceive the real world through a corrected visual layer, demonstrating the potential of XR as an assistive technology for visual accessibility [4].

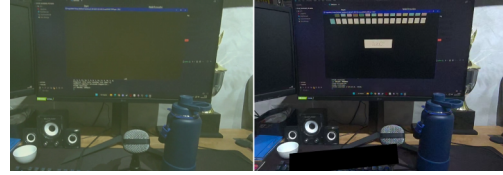


Figure 3: Meta Quest 3 AR scenes: environment with personalized color-vision compensation for a user with mild protanopia on the left, and environment without compensation on the right.

4 CONCLUSION AND FUTURE WORKS

In this work, we presented an XR-based system designed to address the challenges faced by individuals with CVD. By integrating a clinically validated diagnostic tool, the Farnsworth D-15 Arrangement Test, within a controlled VR environment and translating those results into a personalized AR compensation filter, we provide a robust alternative to generic assistive solutions. The transition from VR-based calibration to AR-assisted vision demonstrates how Extended Reality can be leveraged to create highly customized accessibility tools, directly impacting the quality of life and autonomy of the CVD community.

Future works will focus on validating the digital Farnsworth D-15 test using participants with Protan, Deutan, and Tritan deficiencies. It will evaluate the effectiveness of AR filters in improving color discrimination under varied lighting, while collecting feedback on visual comfort and usability to refine interface design and compensation algorithms.

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REFERENCES

- [1] J. Birch. Worldwide prevalence of red-green color deficiency. *Journal of the Optical Society of America A*, 29(3):313–320, 2012. 1
- [2] I. Chaudhari, A. Wahane, U. Rahate, A. Bisen, and A. Gupta. Color-bot: Interactive smartphone assistant for color blindness detection and realtime assistance using llm-based chatbots and machine learning. In *2025 5th International Conference on Soft Computing for Security Applications (ICSCSA)*, pp. 611–616, 2025. doi: 10.1109/ICSCSA66339.2025.11171137 1
- [3] D. Farnsworth. The Farnsworth-Munsell 100-hue and dichotomous tests for color vision. *Journal of the Optical Society of America*, 33(10):568–578, 1943. 1
- [4] D. R. Flatla, K. Reinecke, C. Gutwin, and K. Z. Gajos. Computational trichromacy reconstruction: Empowering the color-vision deficient to recognize colors using augmented reality. In *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, UIST '24. ACM, 2024. doi: 10.1145/3654777.3676415 2
- [5] A. J. Vingrys and P. E. King-Smith. A quantitative scoring technique for panel tests of color vision. *Investigative Ophthalmology & Visual Science*, 29(1):50–63, 1988. 1